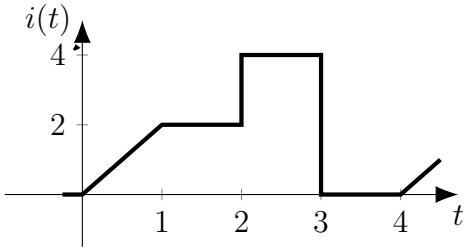
EE2015: Electric Circuits and Networks

Tutorial 9

(October 24, 2025)

1. Consider the periodic signal shown on the right, which represents the current passing through a $10\,\Omega$ resistor.

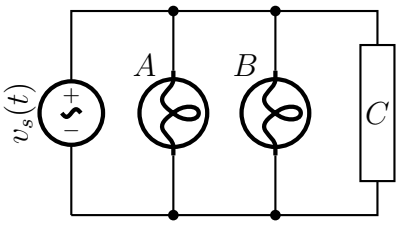


   (a) Find the RMS value of $i(t)$.

   (b) Find the average power absorbed by the resistor.

   (c) Find the value of the energy dissipated in the resistor over one cycle.

2.



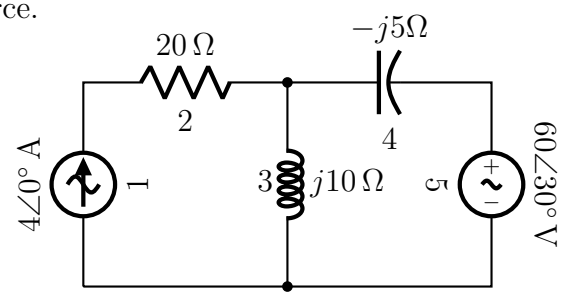
The figure on the left shows two types of household lighting loads that are in parallel to the mains supply of $v_s(t) = 220\sqrt{2}\cos(240\pi t)$ V. Incandescent lamps A and B have *hot* resistances of $101\,\Omega$ and $60\,\Omega$, respectively. The fluouresent lamp C has impedance $\mathbf{Z}(\omega) = 60 + j70\,\Omega$.

   (a) Find the average power consumed by each light.

   (b) Find the average power delivered by the source.

3.

Determine the average power generated by each source and the average power absorbed by each passive element in the circuit shown on the right.



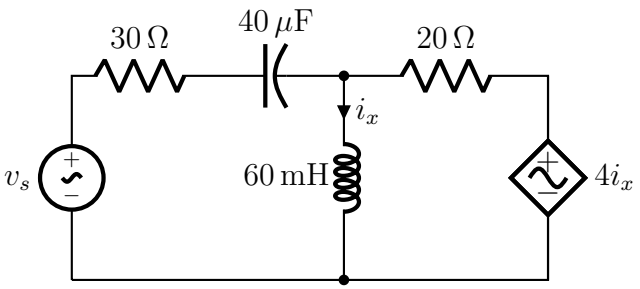
4. A series-connected draws a current $i(t) = 4\cos(100\pi t + 10°)$ A when the applied voltage is $v(t) = 120\cos(100\pi t - 20°)$ V. Find the apparent power and the power factor of the load. Determine the element values that form the series-connected load.

5. Consider a sinusoidal source with value $120\angle 10°$ V that is connected to two load impedances $\mathbf{Z}_1 = 60\angle -30°$ and $\mathbf{Z}_2 = 40\angle 45°$ that are in parallel. Calculate the following quantities: (a) apparent power, (b) real power, (c) reactive power, and (d) power factor.

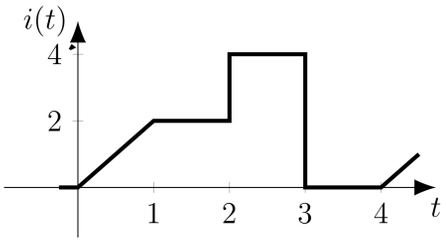
6. When connected to a 220 V (rms), 50 Hz power line, a load absorbs 4 kW at a lagging power factor of 0.8. Find the value of capacitance needed to raise the PF to 0.95.

7.

Find the complex power delivered by $v_s = 100\cos 2000t$ V to the network shown on the right.



1. Consider the periodic signal shown on the right, which represents the current passing through a $10\,\Omega$ resistor.



   (a) Find the RMS value of $i(t)$.

   (b) Find the average power absorbed by the resistor.

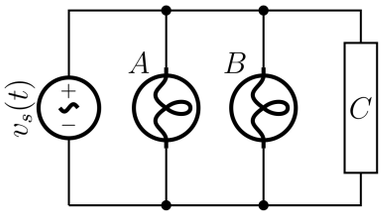
   (c) Find the value of the energy dissipated in the resistor over one cycle.

$$I_{rms}^2 = \frac{\int_0^T [i(t)]^2 dt}{T} = \frac{\int_0^1 4t^2 dt + \int_1^2 4\,dt + \int_2^3 16\,dt}{4} = \frac{4/3 + 20}{4} = 16/3$$

$$\Rightarrow I_{rms} = 4/\sqrt{3}\ A$$

$$P_{avg} = I_{rms}^2 R = 160/3\ W \qquad E = 160/3 \times 4 = 640/3\ J$$

2.



The figure on the left shows two types of household lighting loads that are in parallel to the mains supply of $v_s(t) = 220\sqrt{2}\cos(240\pi t)$ V. Incandescent lamps A and B have *hot* resistances of $101\,\Omega$ and $60\,\Omega$, respectively. The fluouresent lamp C has impedance $\mathbf{Z}(\omega) = 60 + j70\,\Omega$.

   (a) Find the average power consumed by each light.

   (b) Find the average power delivered by the source.

$20\,\Omega$ $\qquad -j5\Omega$

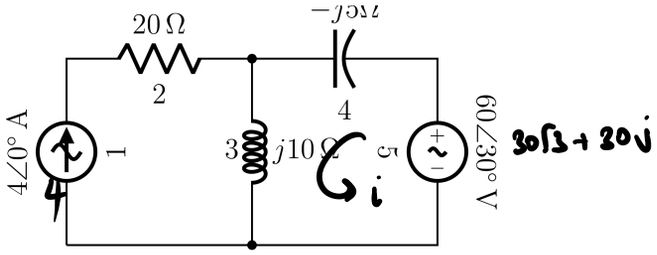
$$V_{rms} = 220V$$

$$P_A = \frac{220^2}{101} = 479.2W \qquad P_B = \frac{220^2}{60} = 808.7W \qquad P_C = \frac{220^2}{8500} \cdot 60 = 341.7W$$

$$P_{avg\ from\ source} = \sum_{j=A,B,C} P_j = 1628W$$

3.

Determine the average power generated by each source and the average power absorbed by each passive element in the circuit shown on the right.



$$30\sqrt{3} + 30j = -5j(i) + 10j(i+4)$$

$$\Rightarrow 30\sqrt{3} + 30j = i(5j) + 40j \Rightarrow (5j)\,i = 30\sqrt{3} - 10j \Rightarrow i = -2 - 6\sqrt{3}j$$

$$P_2 = 16 \times 20/2 = 160W \qquad P_5 = (30\sqrt{3} + 30j)(-2 + 6\sqrt{3}j)/2 = 207.85W$$

$$P_4 = P_3 = 0$$

$$80 + (+2 - 6\sqrt{3}j)(10j) = V = (60\sqrt{3} + 80) + 20j$$

$$P_1 = Re\{VI^*/3\} = 367.8\,W$$

4. A series-connected draws a current $i(t) = 4\cos(100\pi t + 10°)$ A when the applied voltage is $v(t) = 120\cos(100\pi t - 20°)$ V. Find the apparent power and the power factor of the load. Determine the element values that form the series-connected load.

Apparent power $= 120/\sqrt{2} \times 4/\sqrt{2} = 240$ W

power factor $= \cos[-20 - (10)] = \cos[-30] = \sqrt{3}/2$

current leads voltage $\Rightarrow$ capacitive load $\rightarrow$ leading power factor

$V = 120\angle -20°$ $\quad I = 4\angle 10°$

$Z = V/I = 30\angle -30°$

$R = Re(Z) = 30 \times \sqrt{3}/2 = 15\sqrt{3}\,\Omega = 25.98\,\Omega$

$\frac{1}{\omega C} = 30 \times 1/2 = 15 \Rightarrow \frac{1}{100\pi C} = 15 \Rightarrow C = \frac{1}{1500\pi} = 212.21\,\mu F$

5. Consider a sinusoidal source with value $120\angle 10°$ V that is connected to two load impedances $\mathbf{Z}_1 = 60\angle -30°$ and $\mathbf{Z}_2 = 40\angle 45°$ that are in parallel. Calculate the following quantities: (a) apparent power, (b) real power, (c) reactive power, and (d) power factor.

$S = \frac{|V|^2}{Z^*}$

$S_1 = \frac{14400}{60\angle 30} = 240\angle -30°$ $\quad S_2 = \frac{14400}{40\angle -45°} = 360\angle 45°$

$P = S_1 + S_2 = 120\sqrt{3} - 120j + 180\sqrt{2} + 180\sqrt{2}j$

$= [120\sqrt{3} + 180\sqrt{2}] + [180\sqrt{2} - 120]j$

$462.404 + 134.558j$

(a) $\sqrt{462.4^2 + 134.6^2} = 483.3$ W

(b) 462.4 W

(c) 134.6 W

(d) $462.4/483.3 = 0.957$

6. When connected to a 220 V (rms), 50 Hz power line, a load absorbs 4 kW at a lagging power factor of 0.8. Find the value of capacitance needed to raise the PF to 0.95.

$S_1 = (4000 + 3000j)$ W

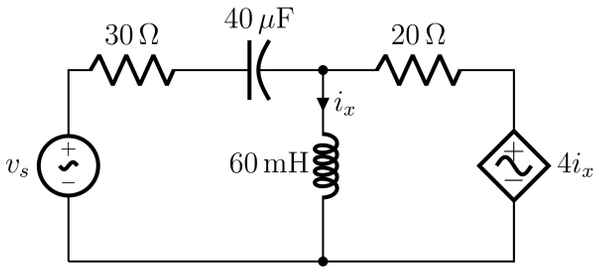
$S_2 = (4000 + 1314j)$ W

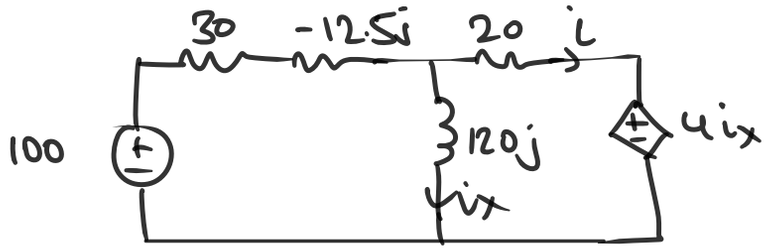
$S_2 - S_1 = -1686j = \frac{220^2}{X_C} = -j\,220^2\omega \cdot C$

$X_C = \frac{1}{j\omega C}$, $\Rightarrow C = \frac{1686}{220^2 \cdot 2\pi \times 50} = 110.89\,\mu C$

7.

Find the complex power delivered by $v_s = 100\cos 2000t$ V to the network shown on the right.





$$100 = (30-12.5j)(i+ix) + 120j\, ix$$

$$100 = (30-12.5j)\, i + (30+107.5j)\, ix$$

$$4ix = -20i + 120j\, ix$$

$$\Rightarrow (1+30j)\, ix = 5i$$

$$100 = (30-12.5j)\frac{(1+30j)}{5} ix + (30+107.5j)\, ix$$

$$ix = 0.105 - 0.309j$$

$$i + ix = \left(\frac{30j-1}{5} + 1\right) ix = (6j+45)\, ix = 1.94 + 0.39j$$

$$P = \frac{1}{2} \times 100\,(1.94 - 0.39j) = 97 - 19.5j$$